KOUSHIKI BHATTACHARYYA

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PERSONAL PROFILE

A dedicated undergraduate student with a strong academic record, currently pursuing B.Tech in Optics & Optoelectronics at the University of Calcutta. My research interests lie in black holes and active galactic nuclei, particularly AGN feedback, the relativistic physics close to the black hole, and its X-ray emission, as well as in extragalactic astrophysics more broadly, including galaxy evolution and the circumgalactic medium. My long-term goal is to contribute to these areas through research in observational & computational astronomy.

EDUCATION

University of Calcutta *Oct 2024 - Present*
B.Tech in Optics & Optoelectronics Engineering, CGPA: 9.32/10.00 (till 4th semester)

Pathfinder Higher Secondary Public School (H.S.)- Class XII *May 2023*
West Bengal Council of Higher Secondary Education, Percentage: 85.2%

Ramkrishna Vivekananda Mission Vidyabhawan- Class X *June 2021*
West Bengal Board of Secondary Education, Percentage: 93.71%

EXPERIENCE

- **Research Intern** *Marseille, France (Remote)*
  Laboratoire d'Astrophysique de Marseille *Oct 2026 - Present*

  Supervisor : Dr. Delphine Porquet
  Project : Observational Study of X-ray Emission of Active Galactic Nuclei

  – Working on observational study of X-ray emission from AGNs, as part of the Galaxies, Étoiles et Cosmologie (GECO) group.

- **Research Intern** *Ann Arbor, USA (Remote)*
  University of Michigan *Sept 2026 - Present*

  Supervisor : Dr. Sayak Dutta
  Project : Probing the CGM of a Foreground Galaxy using a Gravitationally Lensed Quasar with MUSE

  – Analyzing MUSE integral-field data of a lensed quasar behind a foreground galaxy, using the quasar sightlines to probe the galaxy's circumgalactic gas.

  – Deriving the galaxy's properties such as redshift, star formation rate and stellar population age to study how gas flows and star formation drive galactic evolution.

- **Research Intern** *Chandigarh, India (On-site)*
  Council of Scientific and Industrial Research - Central Scientific Instruments Organisation (CSIR-CSIO) *July 2026 - Sept 2026*

  Supervisor : Dr. Neha Khatri
  Project : Optical Design of a Snapshot Hyperspectral Camera using CASSI

  – Designed and optimized the optical module of a CASSI-based snapshot hyperspectral camera for a 6U CubeSat Earth-observation mission in ∼500 km LEO, covering the 400–900 nm VNIR spectral range, using Zemax OpticStudio. Analyzed optical aberrations, image quality, and system performance for the hyperspectral imaging system.

- **Undergraduate Summer Research Intern** *Warsaw, Poland (Remote)*
  Centrum Astronomiczne im. Mikołaja Kopernika PAN (NCAC PAS) *June 2026 - Aug 2026*

Supervisor : Mr. Parikshit Partha Biswas
Project : Relativistic Broadening of AGN Emission with General Relativity Ray-Tracing Code GYOTO

- Performed general relativistic ray-tracing simulations of X-ray emission from active galactic nuclei in Kerr spacetime using GYOTO, with radiative transfer outputs from the TITAN photoionization code as inputs to model relativistically broadened Fe Kα emission.
- Executed and managed large-scale GYOTO simulations on a Linux-based high-performance computing cluster using SLURM.
- Analyzed the effects of black hole spin, observer inclination, and higher-order photon trajectories on simulated spectra and images using Python.

PROJECTS

Assembling a Fabry-Perot Interferometer using a Laser Source *Kolkata, West Bengal*
University of Calcutta *Jan 2026*

- Set up and precisely aligned a Fabry–Perot interferometer using a coherent laser source and CCD-based detection system to digitally record interference fringes.
- Gained hands-on experience in optical alignment, cavity adjustment and basic fringe analysis through computer-based imaging.

TECHNICAL & COMMUNICATION SKILLS

Softwares & Tools :	MATLAB, Zemax OpticStudio, SLURM, LaTeX
Programming :	Python, C/C++
Libraries :	Scipy, Numpy, Astropy, Matplotlib
Scientific Codes :	GYOTO
Soft Skills :	Problem solving, adaptability, and a strong willingness to learn
Languages :	English, Bengali, Hindi

SELECTED COURSEWORK

Classical Mechanics, Quantum Mechanics (Introductory), Electromagnetism, Foundation of Optical Science, Fundamentals of Optoelectronics, Network Theory & Circuit Analysis, Digital & Analog Electronics & Communication, Engineering Mathematics (I, II & III), Image Science, LASER Physics, Radiometry and Photometry, Signals & Systems, Fourier Optics & Holography, Digital Image Processing, Gaussian Beam Optics & LASER Applications.

AWARDS & ACHIEVEMENTS

- **Secured 1st place in inter-college quiz competition on Optics & Astronomy** organized by student chapters of SPIE, OSI & OPTICA, University of Calcutta. *May 2025*
- Recipient of the **Jagadish Bose National Science Talent Search Scholarship (Junior)**. *Dec 2021*

ACTIVITIES

- **Volunteer** *May 2026 - Present*
  OSI – Optical Society of India (University of Calcutta Student Chapter)
- **Secretary** *Apr 2026 - Present*
  SPIE – The International Society for Optics and Photonics (University of Calcutta Student Chapter)
  - Assisting in the planning and organization of outreach programs, seminars, and chapter activities.
- **Student demonstrator** *Aug 2025*
  Jigyasa Programme (CSIR-KVS Student-Scientist Connect Initiative)
  - Explained and demonstrated scientific and optical models to school students as part of a science outreach initiative.